

Opportunity Dataset — Data Cleaning Documentation

Overview

- Raw file: Copy_of__Opportunity_Data_-_Sheet1.csv
- Raw shape: 5,733 rows x 33 columns
- Cleaned shape: 5730 rows x 32 columns
- Grain: one row = one Opportunity (Internship, Course, Career, Competition, etc.)

Issues Found and How They Were Fixed

Structurally corrupted rows (3 rows): unescaped JSON in tracking_questions broke the CSV column alignment. Dropped as unrecoverable.

Invalid category values: turned out to be the same 3 corrupted rows - resolved automatically once those rows were dropped.

Location inconsistency: casing/spelling variants (virtual/Virtual/vitruual, wfm/WFM/work from home) collapsed into 2 clean values. Also found and fixed a hidden bug where "Null" (title case) slipped past our first missing-value cleanup because it only checked for "NULL"/"null" exactly.

Duration_type inconsistency: 17 variants (week/weeks/yearssss/etc.) mapped down to 6 standard units. One ambiguous value ("da") was resolved using context (duration=1, event-type opportunity) rather than guessed blindly.

Role inconsistency: 429 unique values, mostly placeholder text ("role") and gibberish test entries. Used a whitelist strategy - mapped known real roles, nulled everything else, rather than trying to clean each junk value.

Currency_type inconsistency: EURO merged into EUR.

Wrong data types: created_at/modified_at/last_date_to_apply converted from epoch-milliseconds to real dates (one fake 1970 placeholder date fixed). fee/duration/microscholarship converted to numeric (one 6-trillion fake fee, tied to a record literally named "Testing auto approve", set to missing). is_archived/is_auto_approve converted to real booleans.

Text quality: HTML tags stripped from description fields (11 rows). Broken apostrophes repaired, e.g. "today?s" -> "today's" (7 rows).

Test/QA data contamination: 397 rows (6.9% of the dataset) identified as internal test entries via name pattern matching and empty-core-field detection. Flagged with is_likely_test_data rather than deleted, since deletion risks losing real data if the heuristic is imperfect.

Nested JSON relationship columns: 9 columns (Badge, Cohort, Eligibility, Reward, Testimonial, etc.) held raw JSON pointing to other tables - not usable for direct analysis. Converted to has_x boolean flags or num_x counts. Original JSON preserved in opportunity_raw_json_reference.csv, keyed by opportunity_id.

Low-value columns: pk (constant after row cleanup) and current_editor (96% missing, internal admin metadata) dropped.

Known Limitations

- is_likely_test_data is a heuristic, not exhaustive - some Lorem-Ipsum-style test rows may still be uncaught.
- is_archived is left as missing/unknown for 90% of rows rather than assumed False - a documented choice, not a certainty.
- Mojibake repair only targets the apostrophe pattern; other corruption types weren't systematically searched for.